Decoding the neural correlates of dream recall from EEG using machine learning and Riemannian geometry

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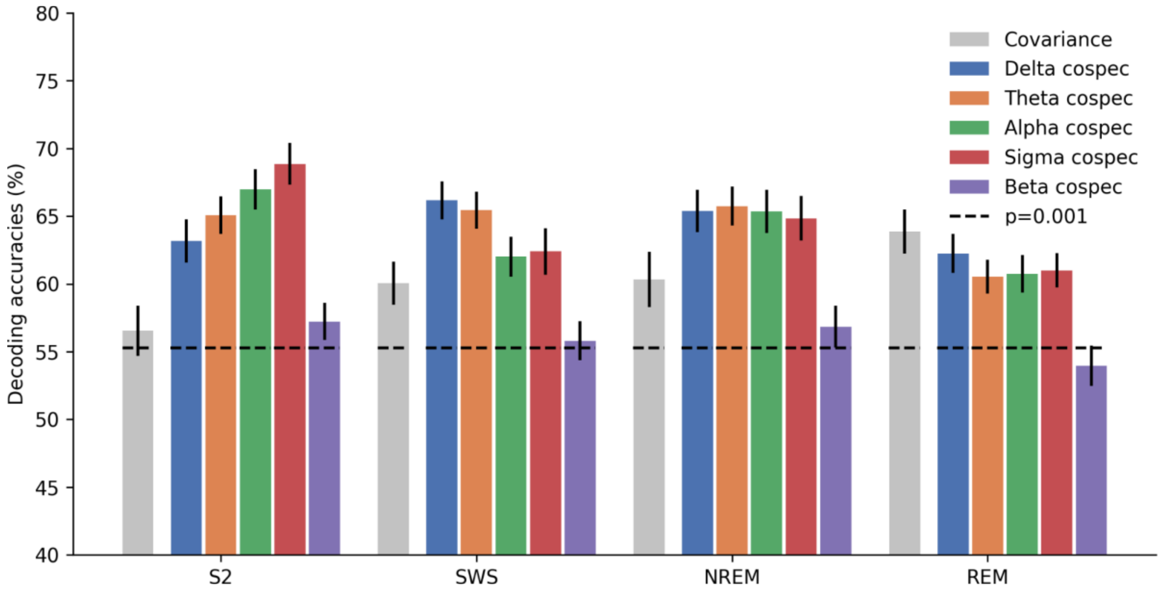
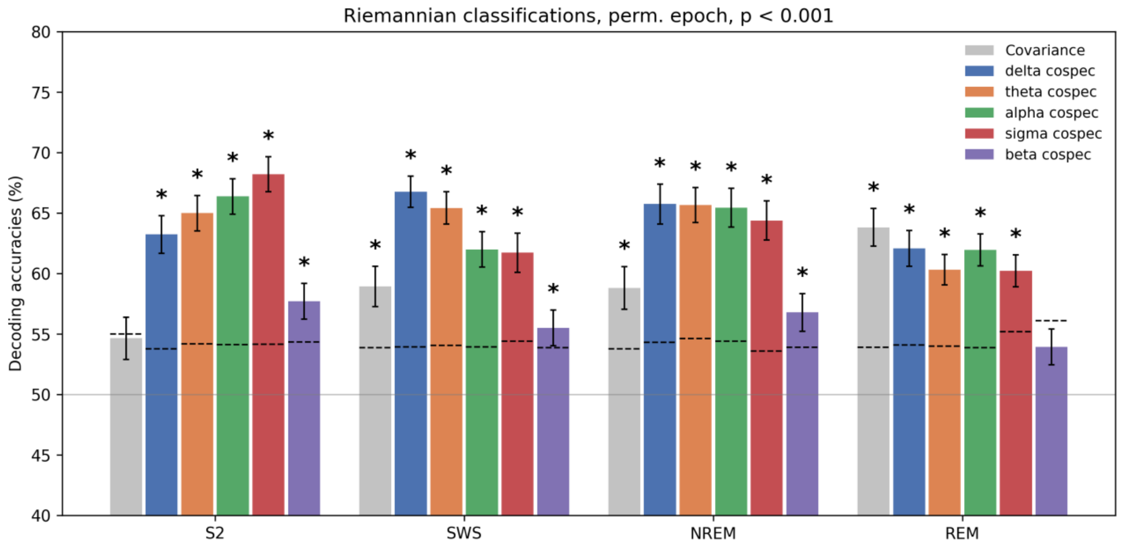
+ Alexandre Louis

Riemannian decoding: replication of Fig. 2



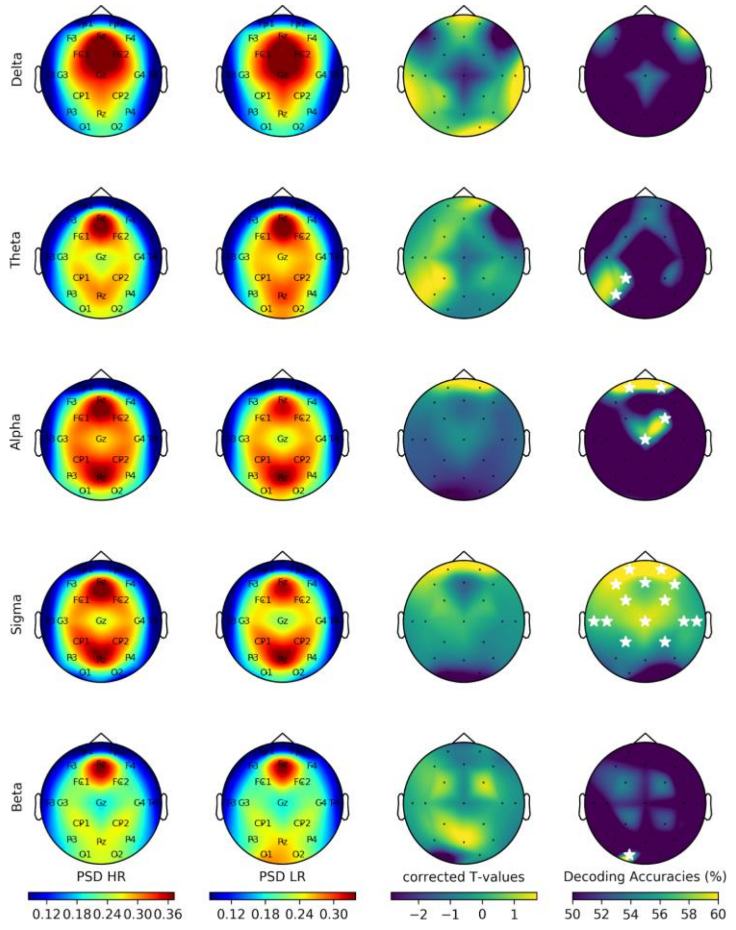
Arthur's pipeline



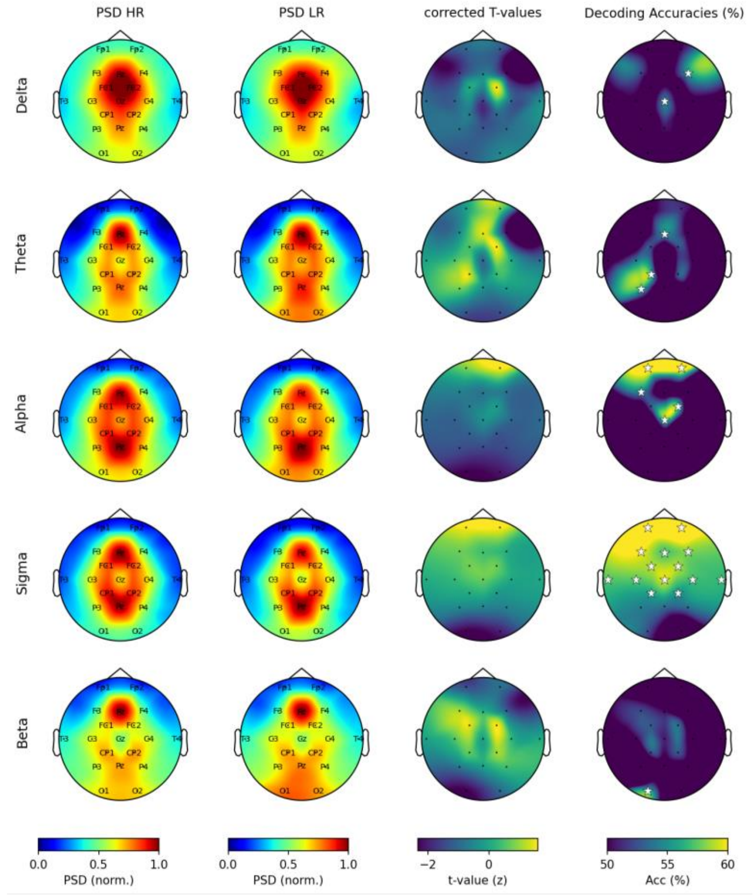
Replication

My null distribution is computed by drawing a new epoch subsample at each permutation, whereas Arthur permutes the labels on a single fixed subsample. My null therefore also incorporates the subsampling variance, which widens it and scatters the thresholds instead of aligning them.

PSD topomaps, replicating Fig. 3

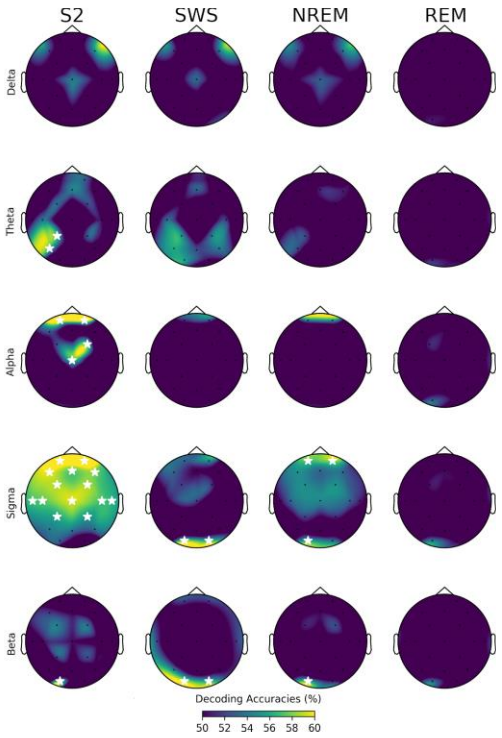


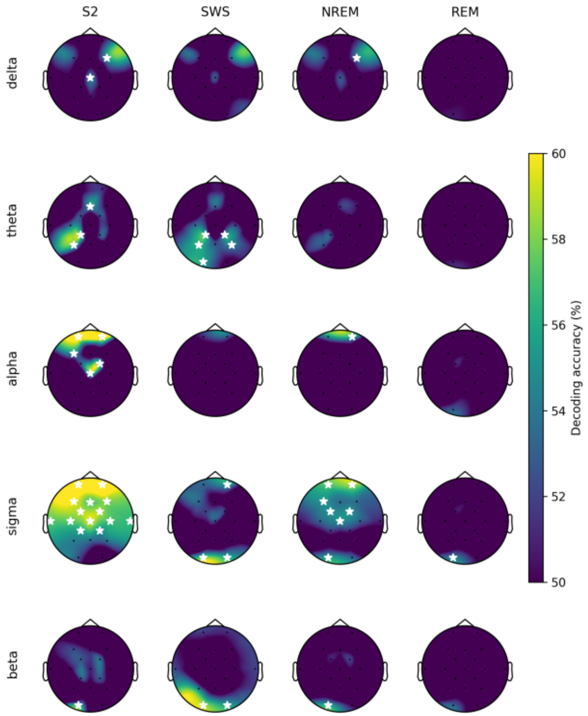
Arthur's pipeline



Replication

LDA decoding topomaps, replicating Fig. 4

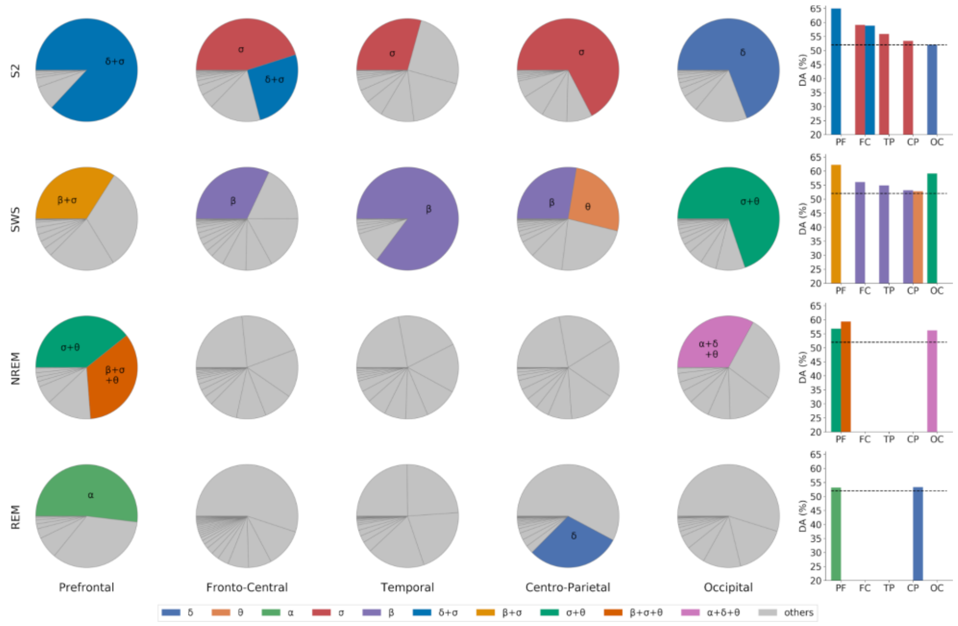


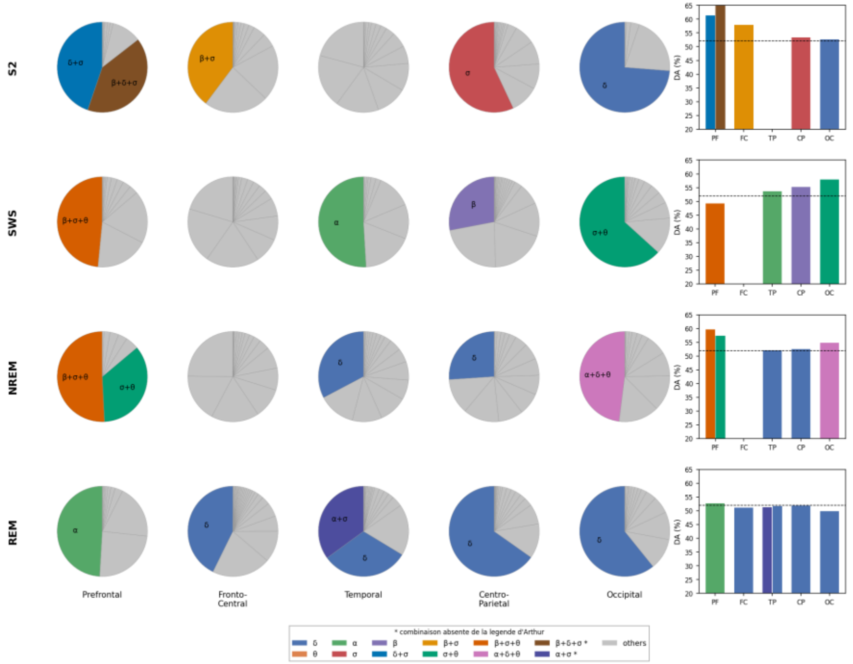


Same background (per-electrode decoding accuracy); the significance criterion differs: Arthur flags an electrode via an analytical binomial threshold; I use a permutation test.

Note: i could not fully reproduce his exact star map because the threshold depends on N (epoch count from his info_data.csv), a file i don't have.

Fig. 5, partial replication



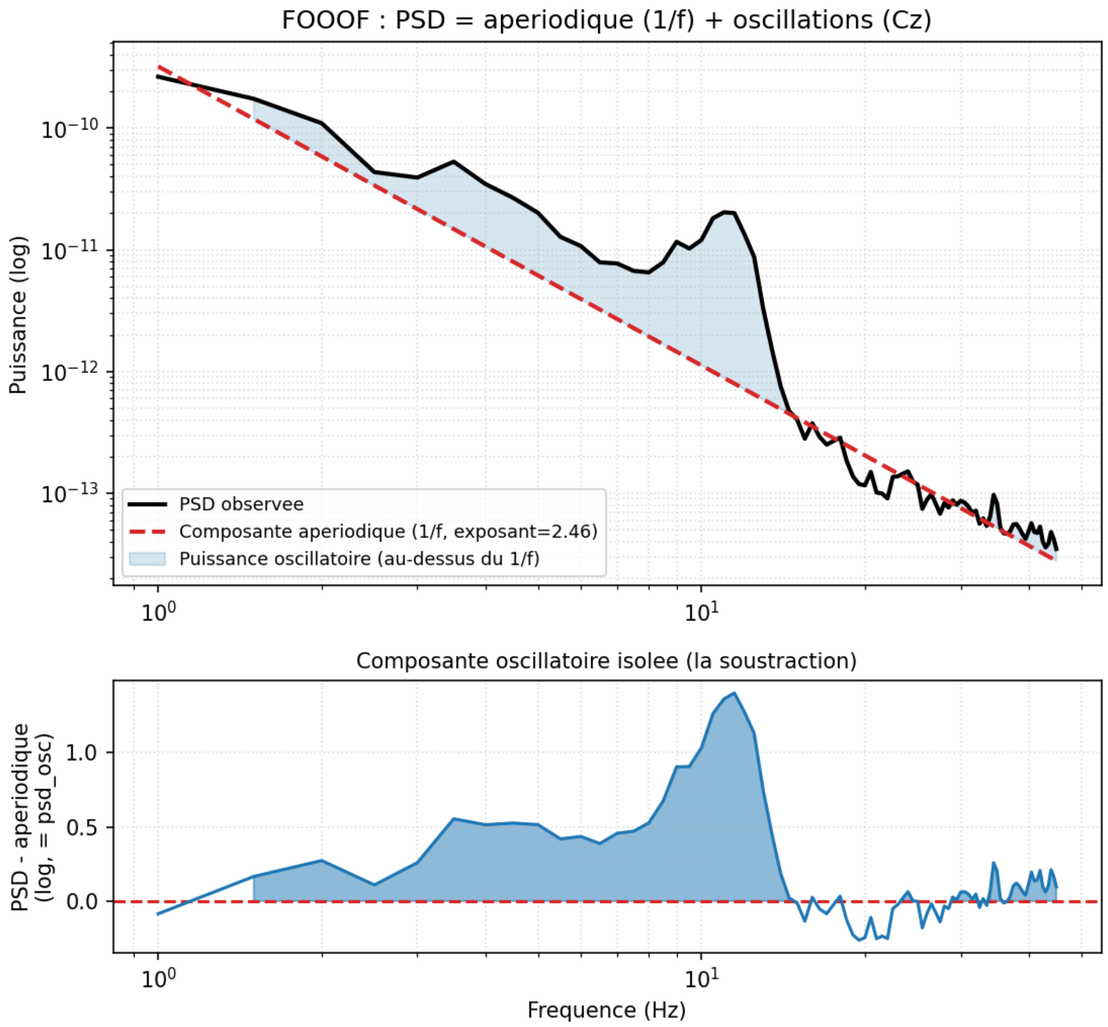


- *REGIONS (electrode-to-ROI mapping) imported from params.py, added to .gitignore, never committed. Our ROIs are not the same as theirs.*
- *The public code divides each band into 5 parts. The thesis version (combinations, gray) comes from unpublished code: our figure is a reconstruction.*
- *His dotted line is not a permutational threshold but an analytical binomial threshold, binom.isf(0.001, N, 0.5)/N, where N = the pool of epochs per stage (5,887 to 21,588), corresponding to 51–52%. This calculation treats each epoch as an independent trial, even though they are nested within 36 subjects. As a rough guide, an N at the subject level would place the threshold around 75%.*
- *Two combinations not included in his legend stand out in our analysis, including β+δ+σ in the prefrontal S2 at 66.78%, our best accuracy, which had not been observed until now.*

New features

FOOOF :

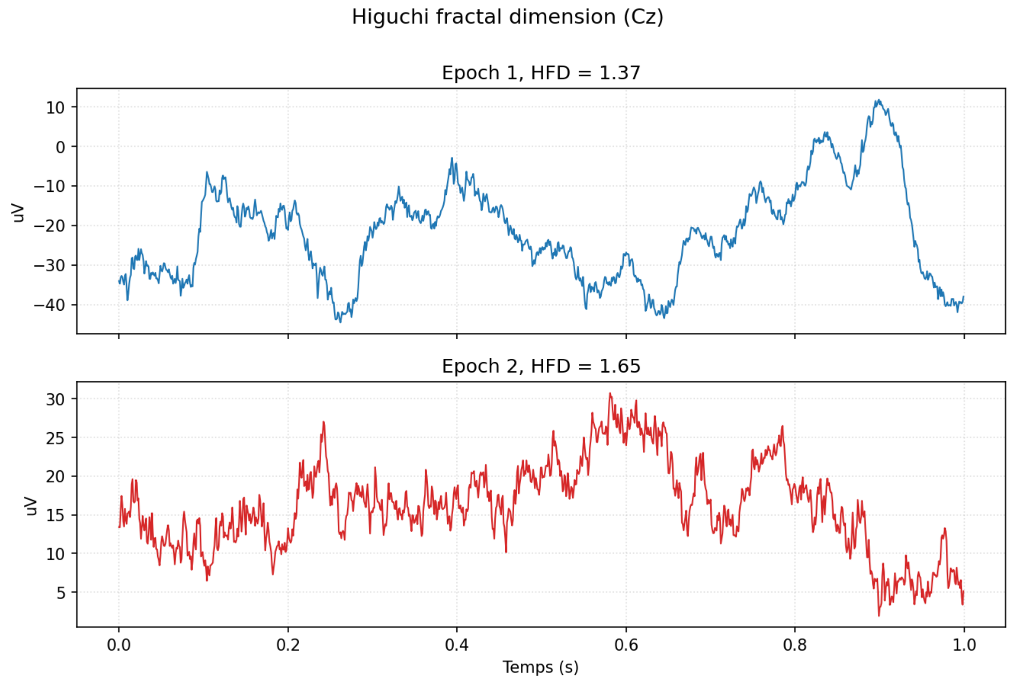
- Oscillatory power ratio on each band
- Aperiodic spectral exponent

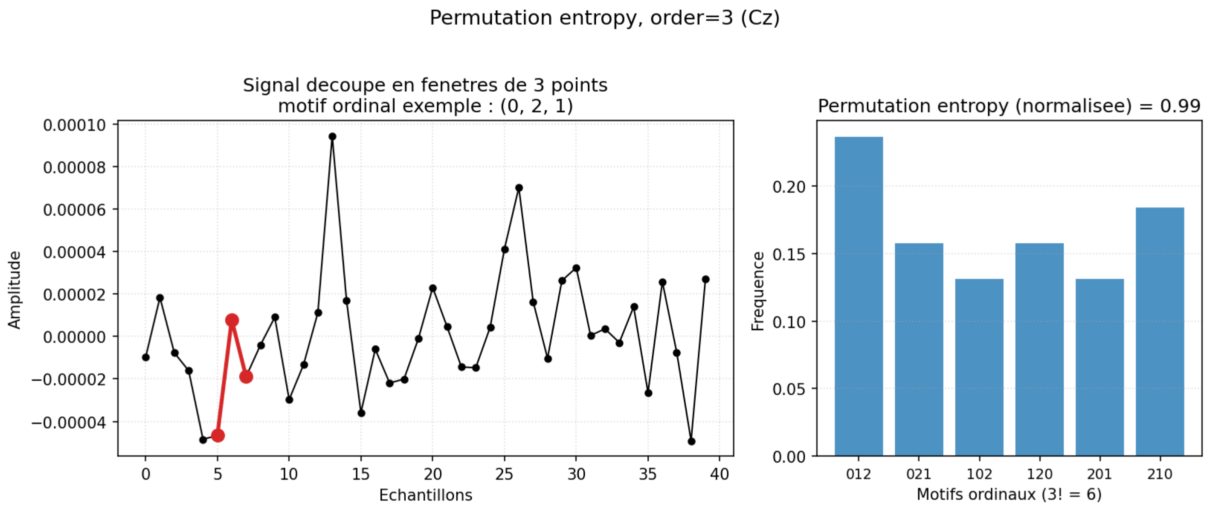


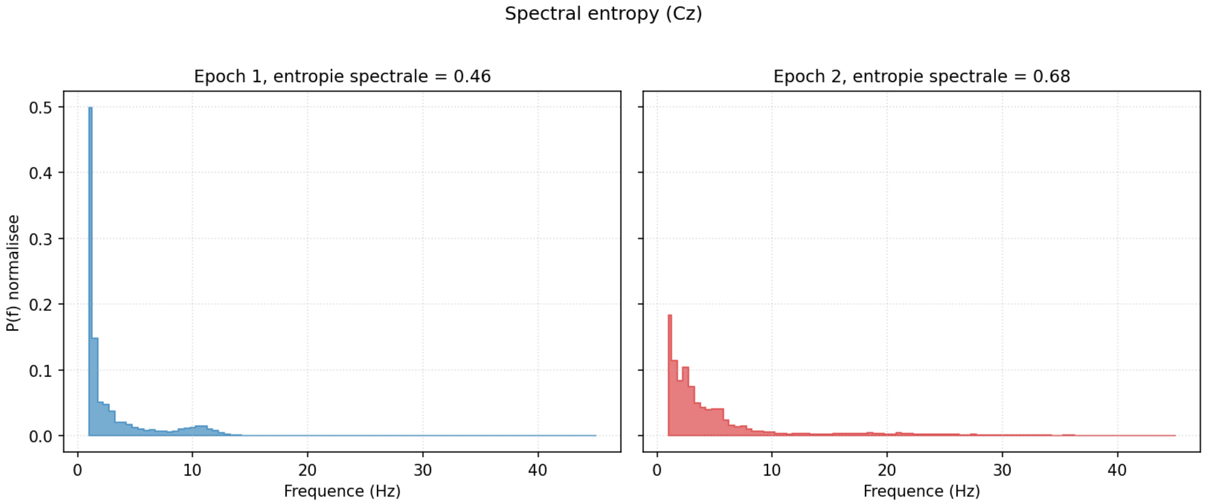
New features

Complexity :

- Permutation entropy (default value = order 3)
- Higuchi fractal dimension (default k_{max}= 10)
- Spectral entropy







Adding overlap

- **Role of overlap**: stabilize the estimators (better-conditioned SPD matrices, less noisy bootstrap), not gain fictitious statistical power

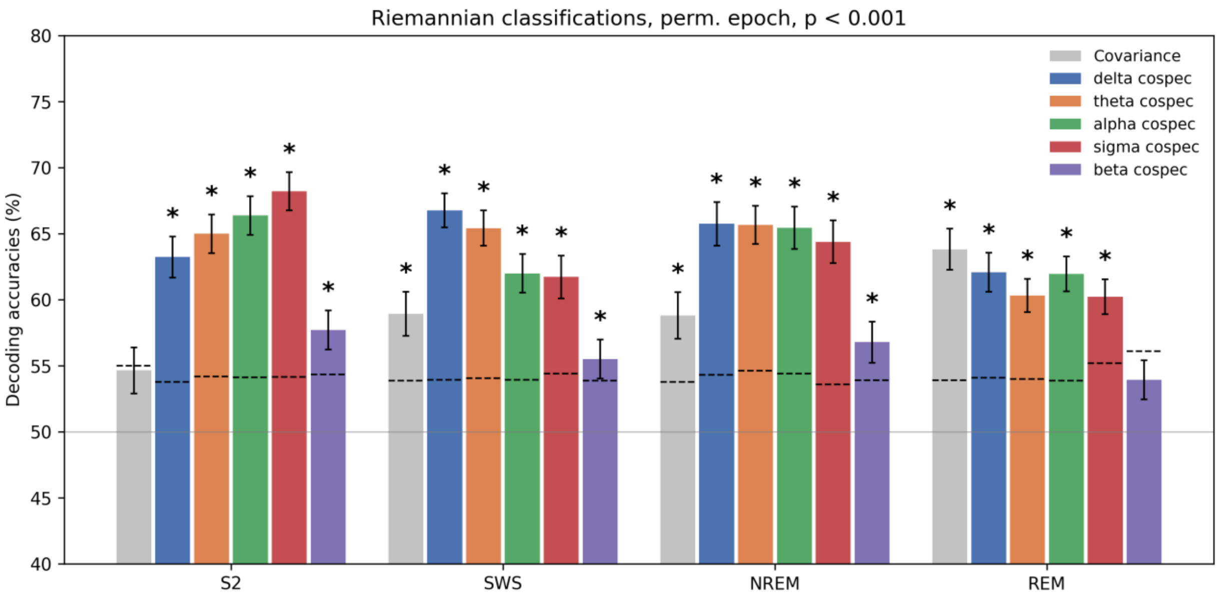
- **Overlap stays confined within a subject**: since a CV fold leaves out whole subjects (LPGO), a subject's overlapping epochs are always in the same fold together, never split between the training subjects and the held-out test subjects.

- **No data leakage**: the train/test split is done **per subject** (Leave-P-Subjects-Out), not per epoch, so all of a subject's epochs, including overlapping ones, fall on the same side of the split.

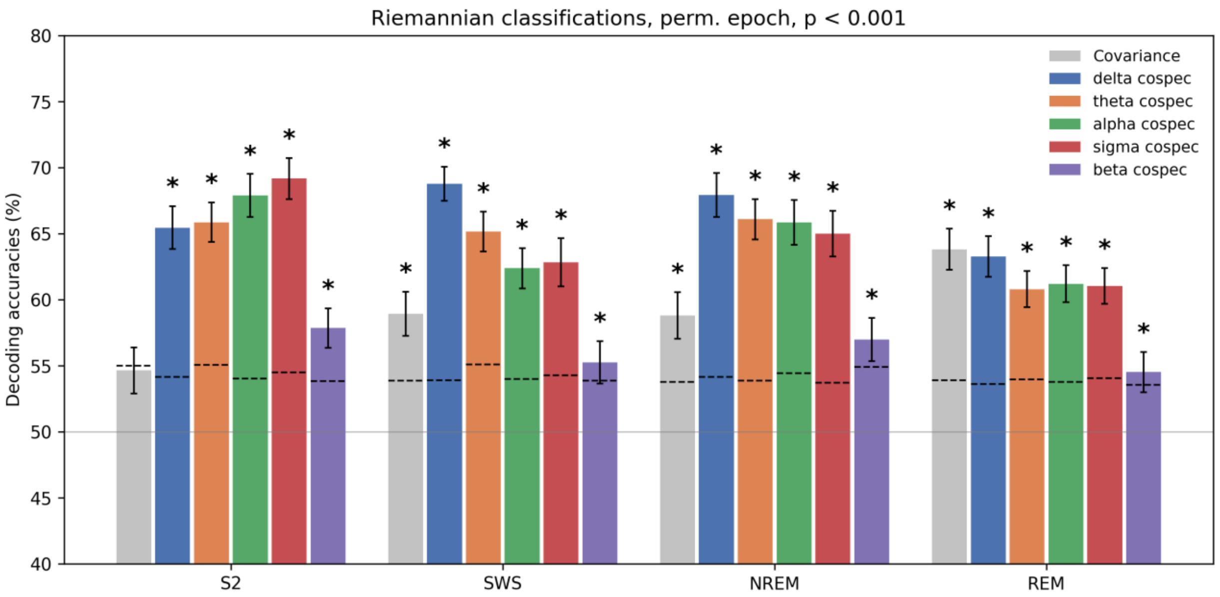
```
WINDOW         = 1000

OVERLAP        = 500
OVERLAP_COSP   = 0.75
```

Adding overlap



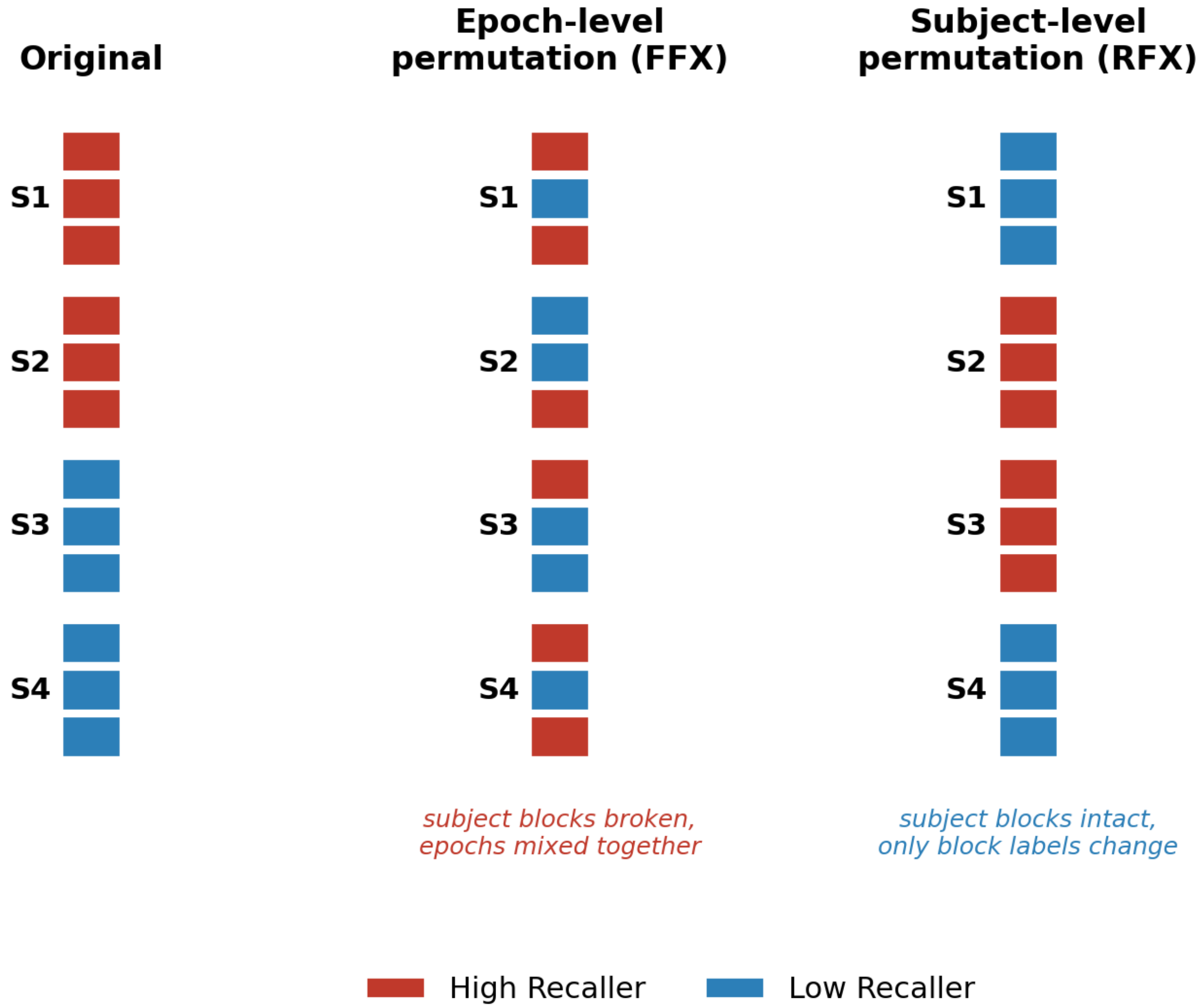
Without



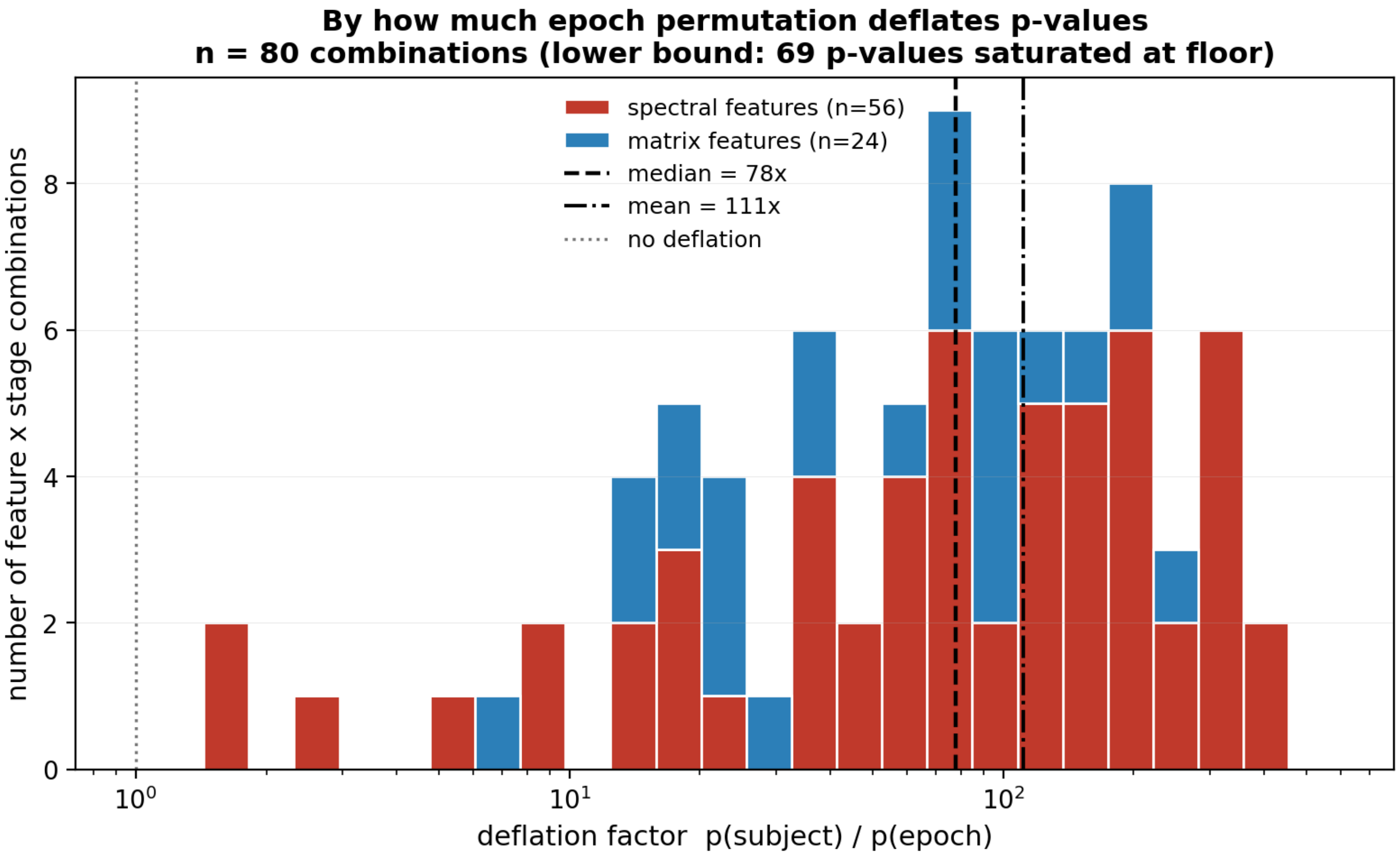
With overlap

Overlap yields a mean gain of +0.62 pts, median +0.46, with 17/24 combos improving. The gain concentrates on cosp_delta (+2.0 to +2.2 pts across all stages), and to a lesser extent the S2 cospectra.

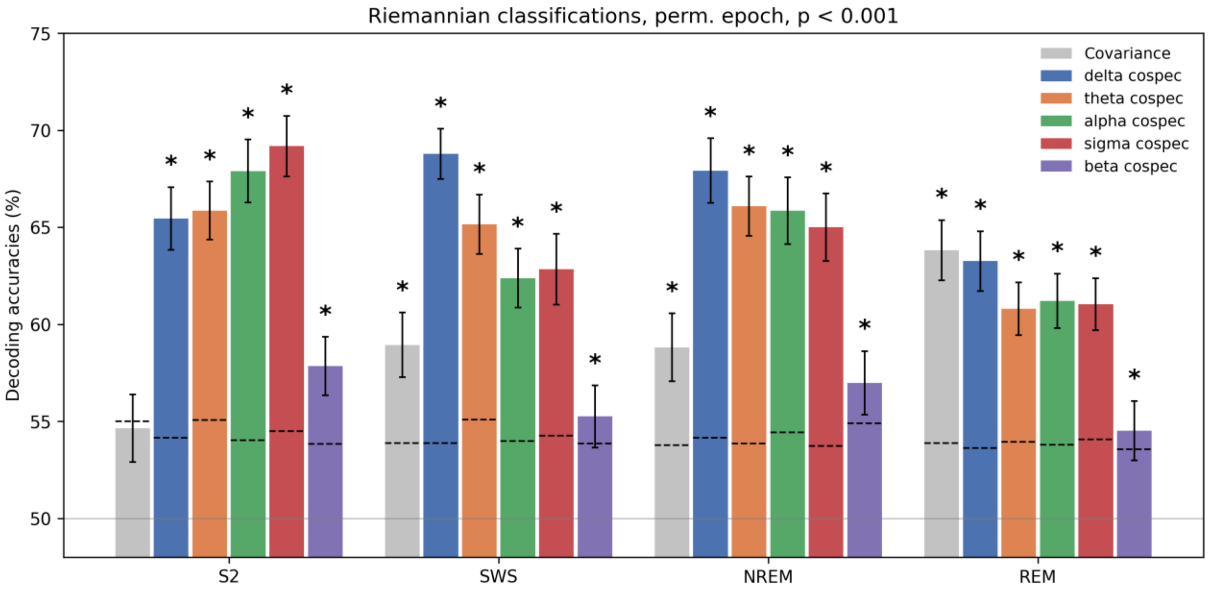
A new way to compute the permutations

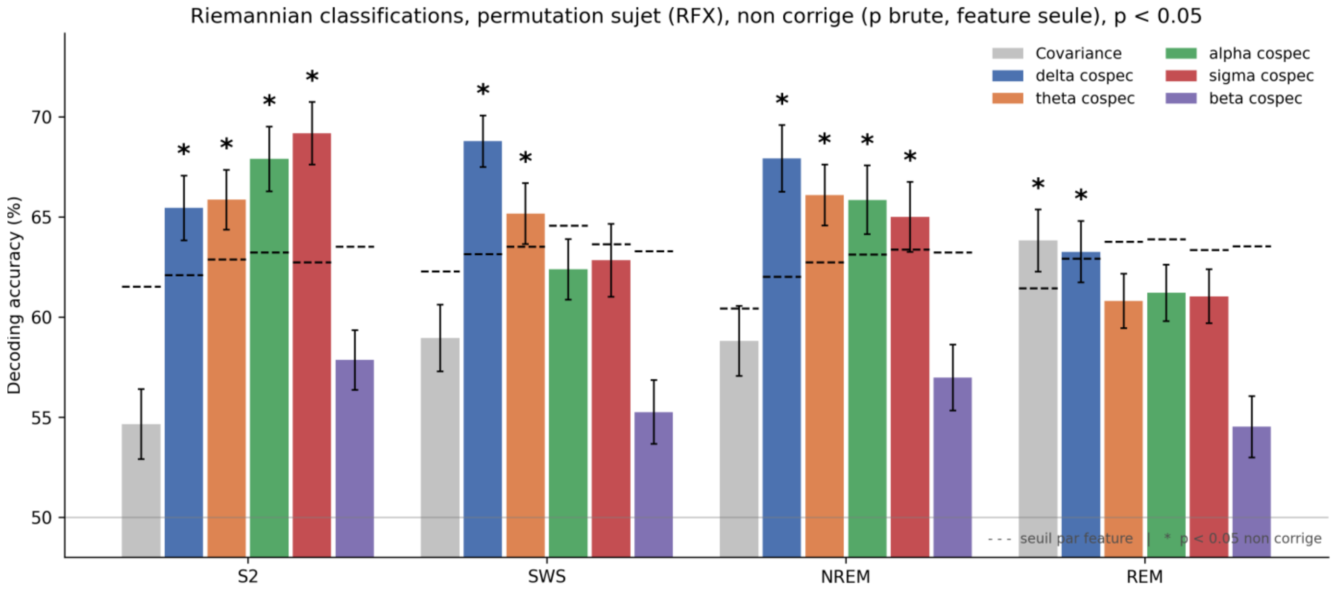


A new way to compute the permutations



Results after new perms

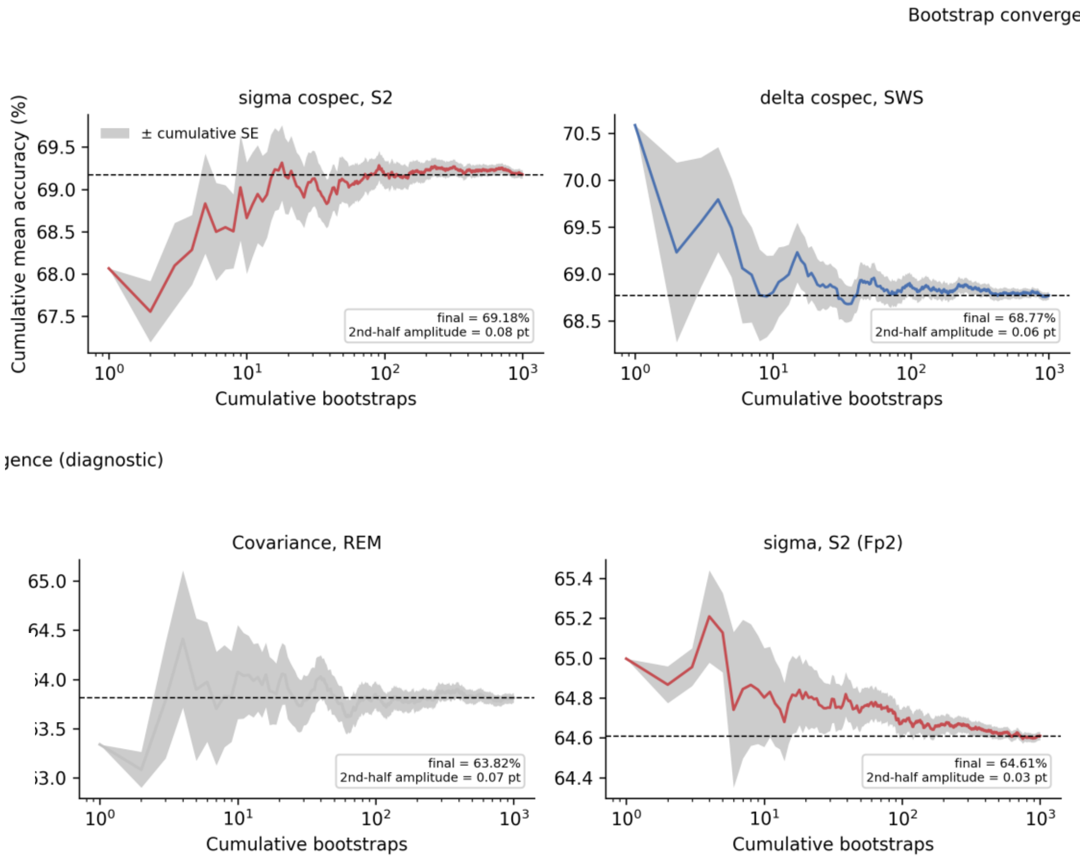


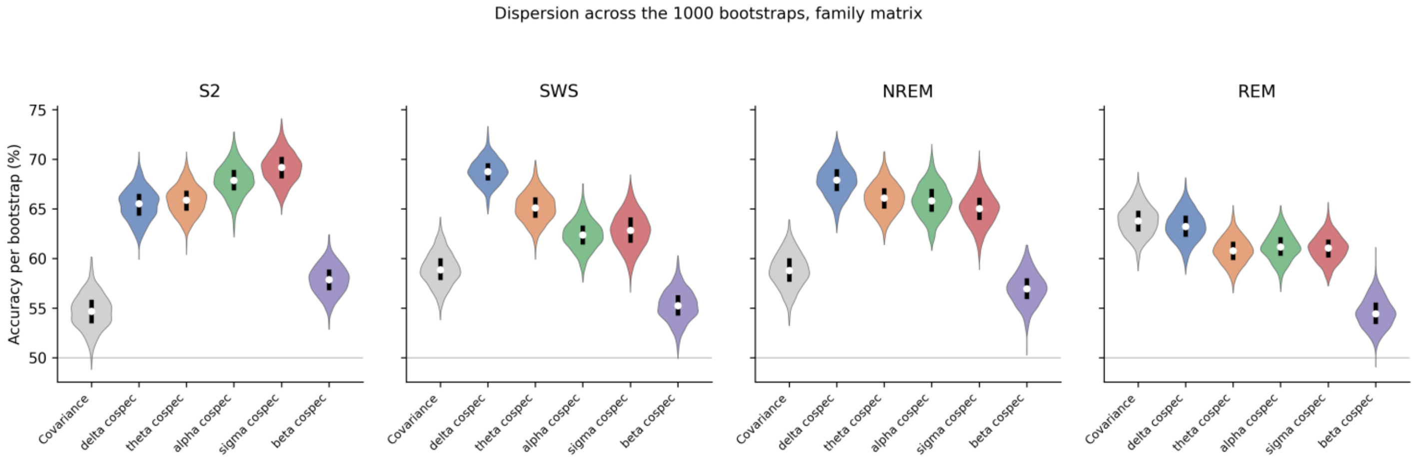


Stability of the Bootstrap Estimate

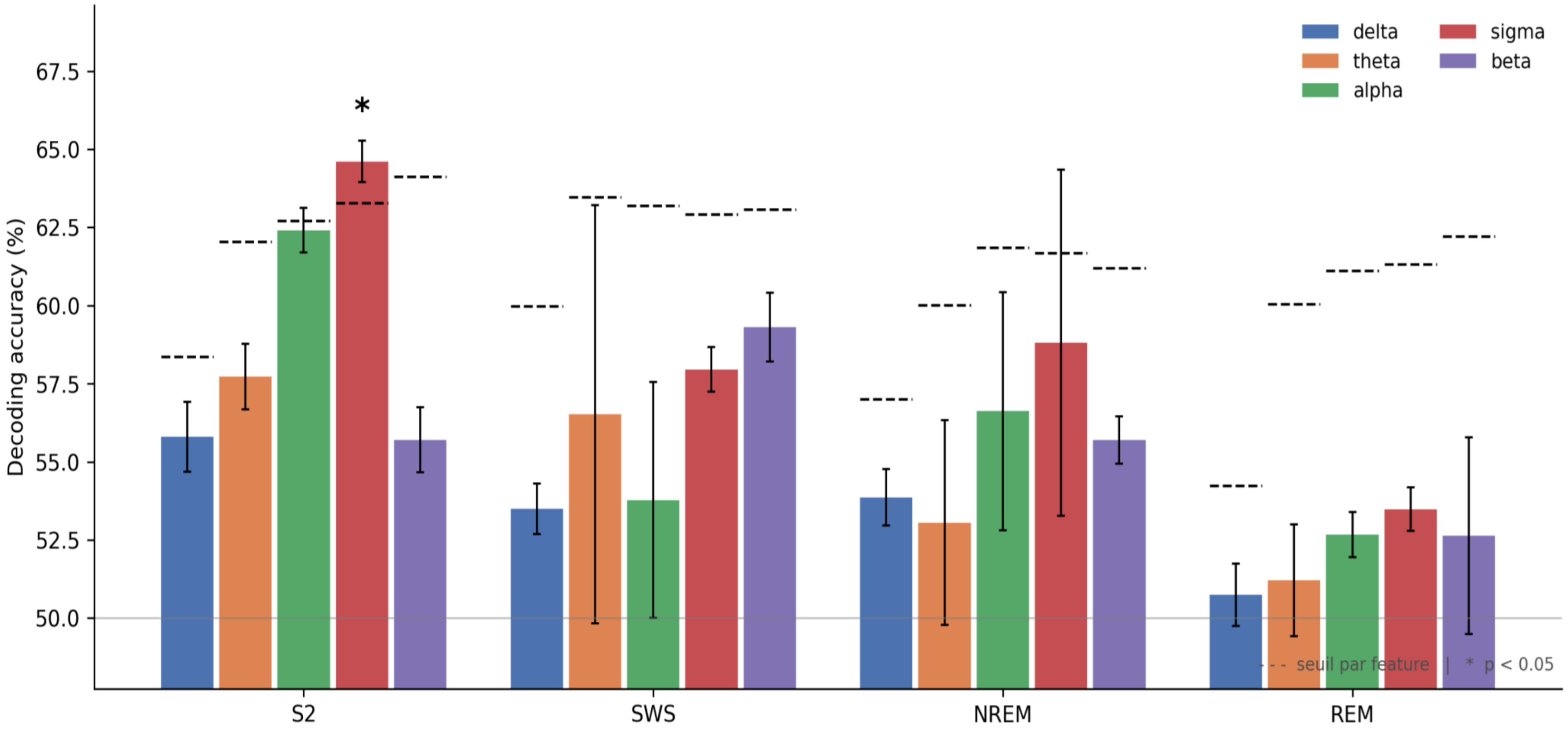
- **Convergence**: The average accuracy stabilizes well before 1,000 runs; the number of bootstrap samples does not affect the results.

- **Dispersion**: The variability across feature/stage combinations justifies the error bars and confirms that the significant effects are not due to a favorable draw.

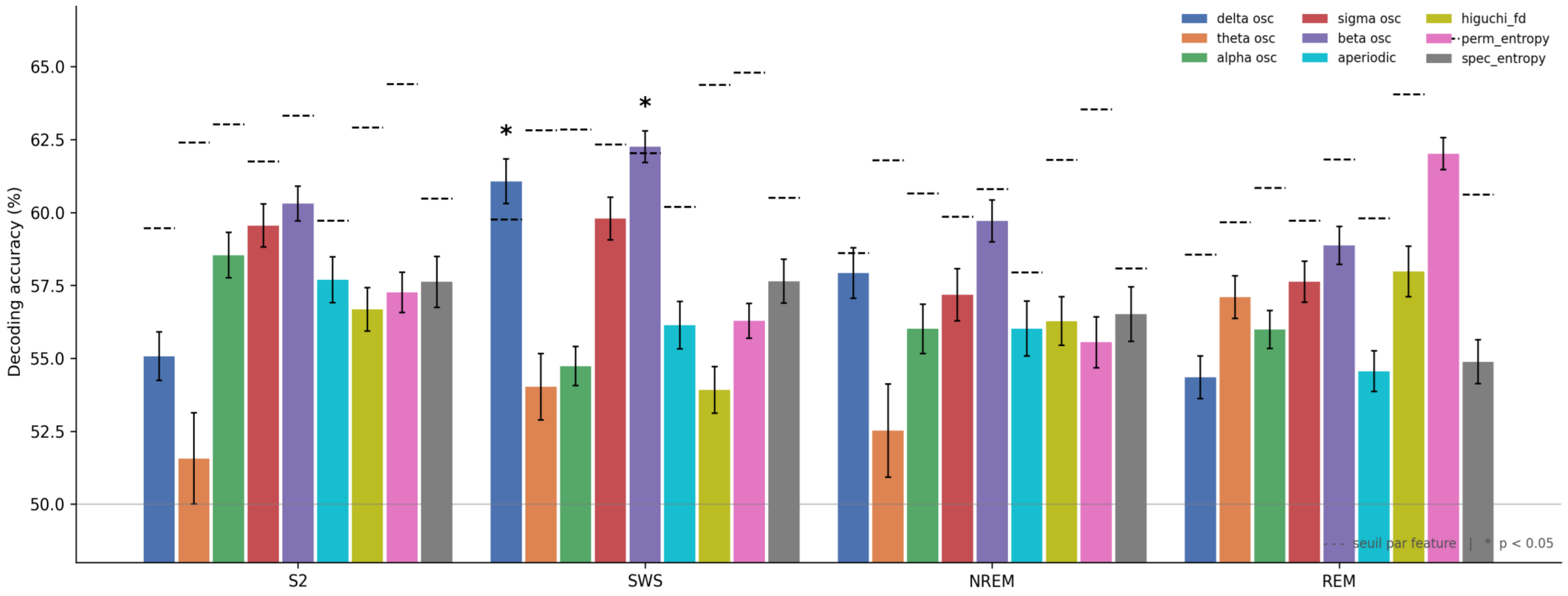




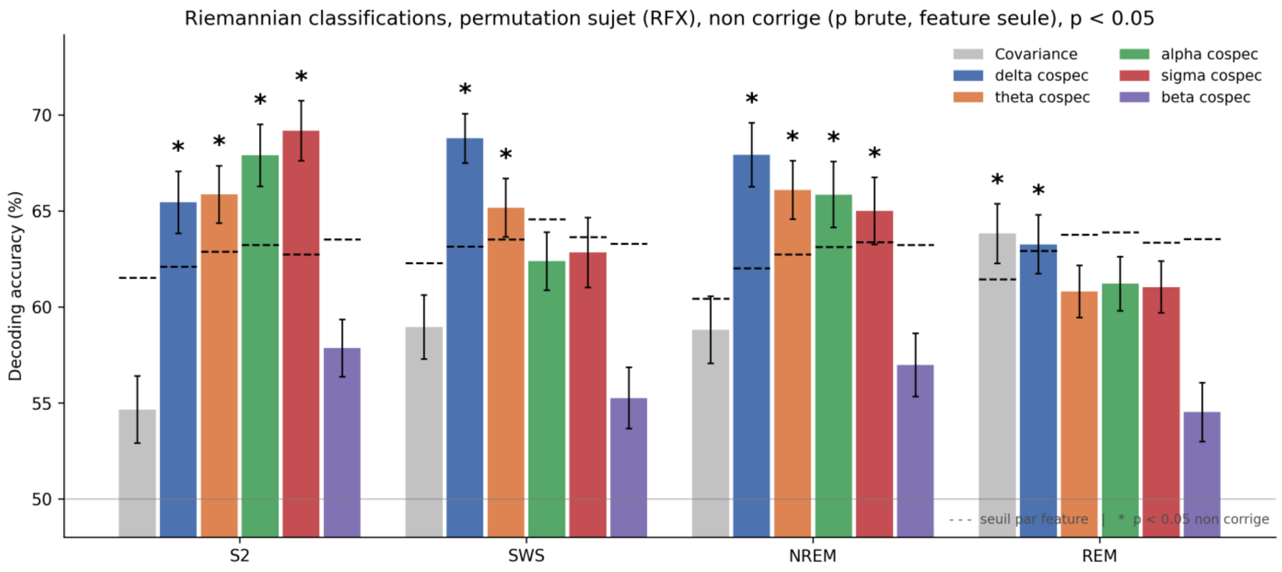
Maxstats: across the 19 électrodes : PSD

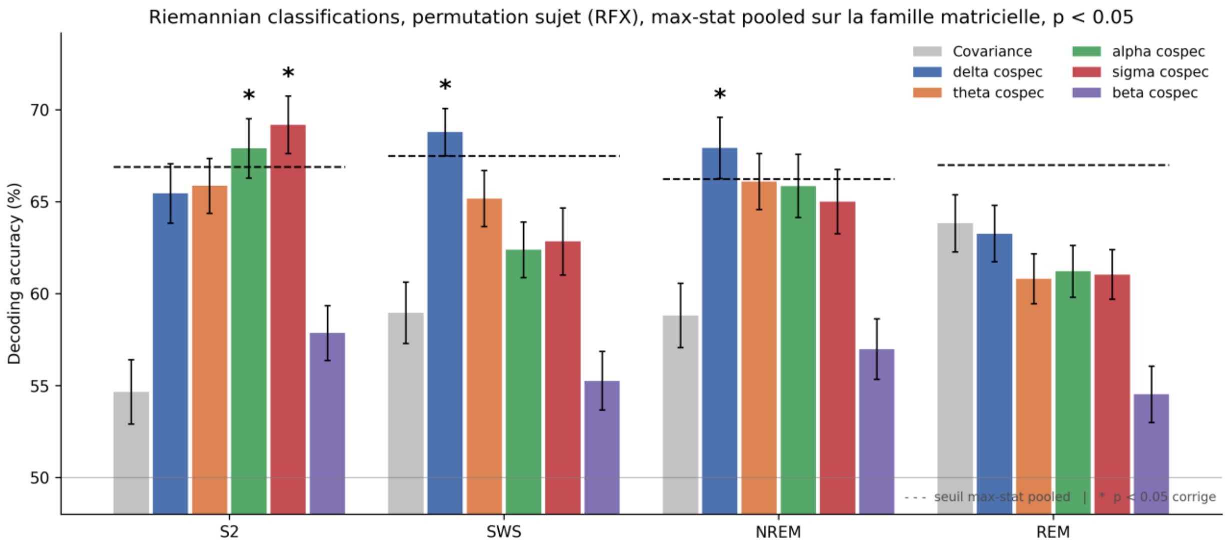


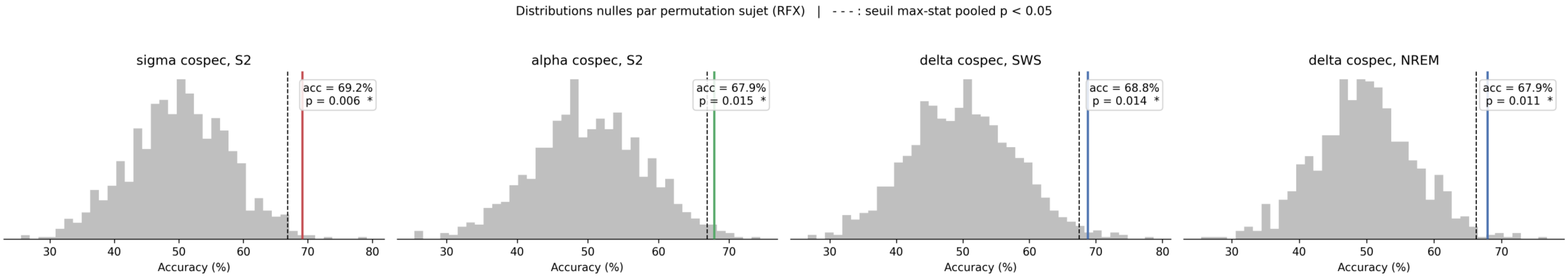
And our new features



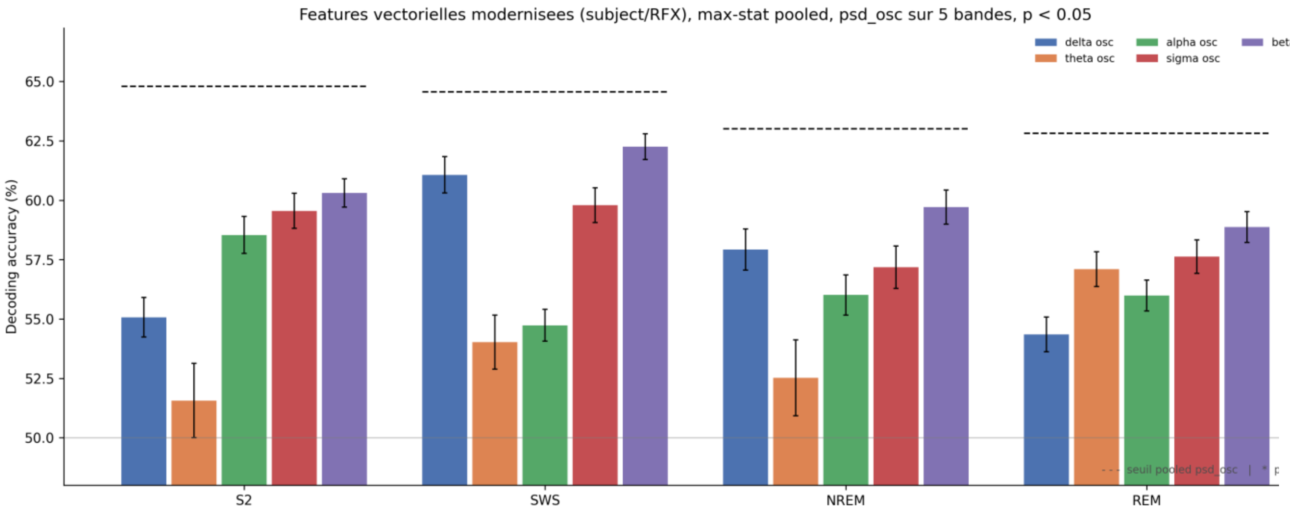
Maxstats: across all feature × stage combinations

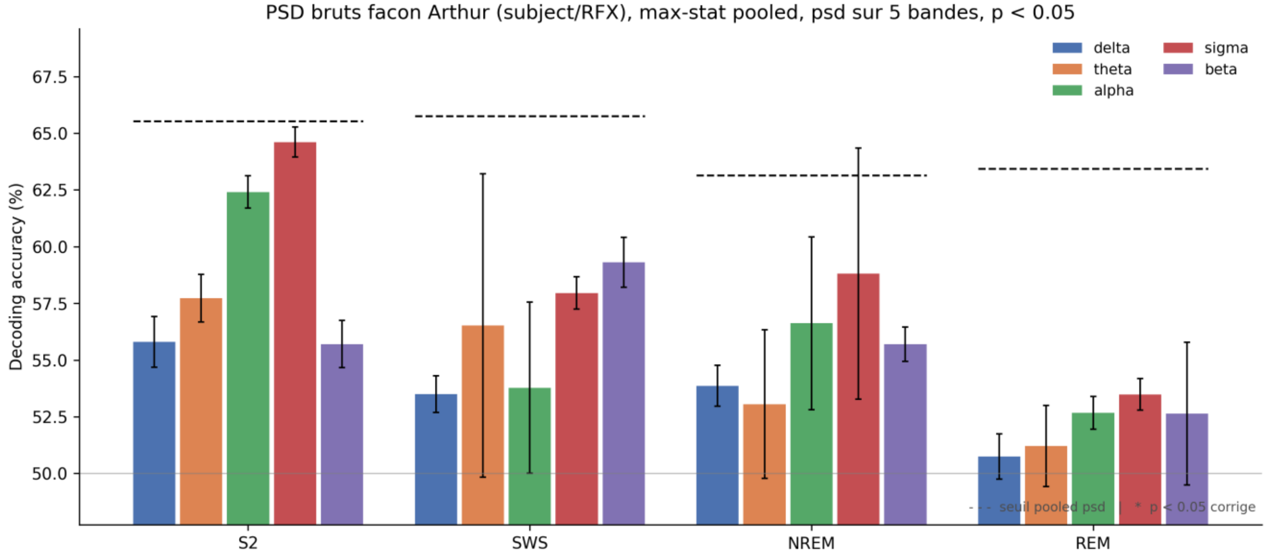


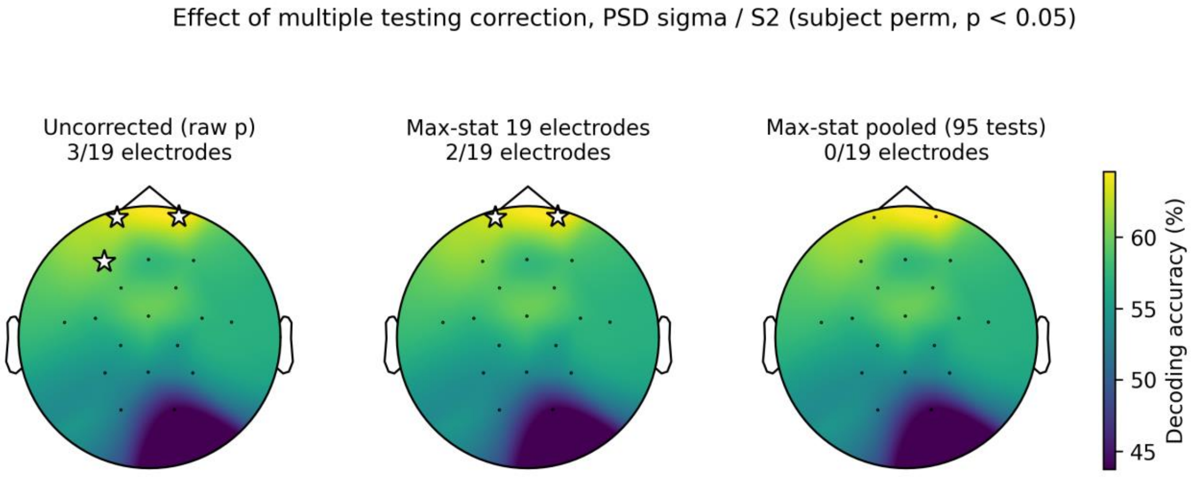




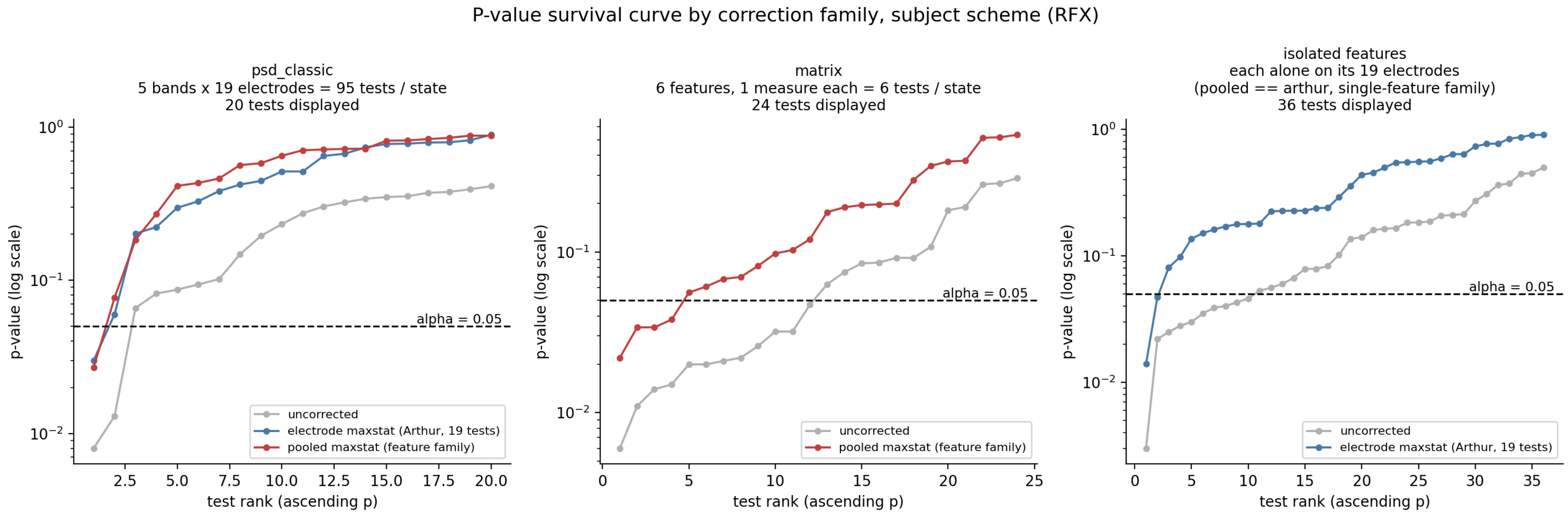
Osc power ratio (fooof) & PSD



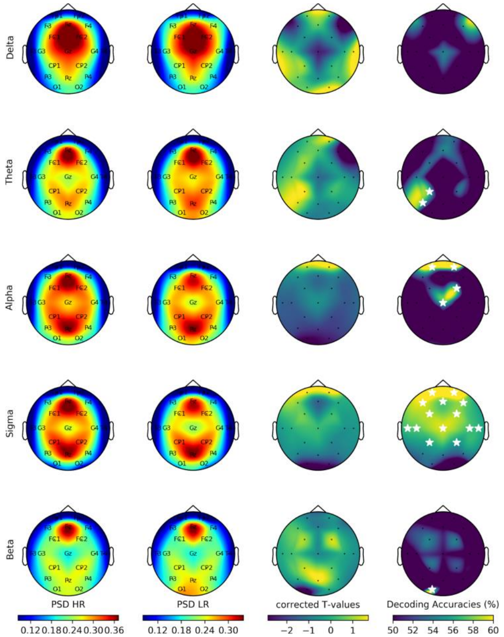


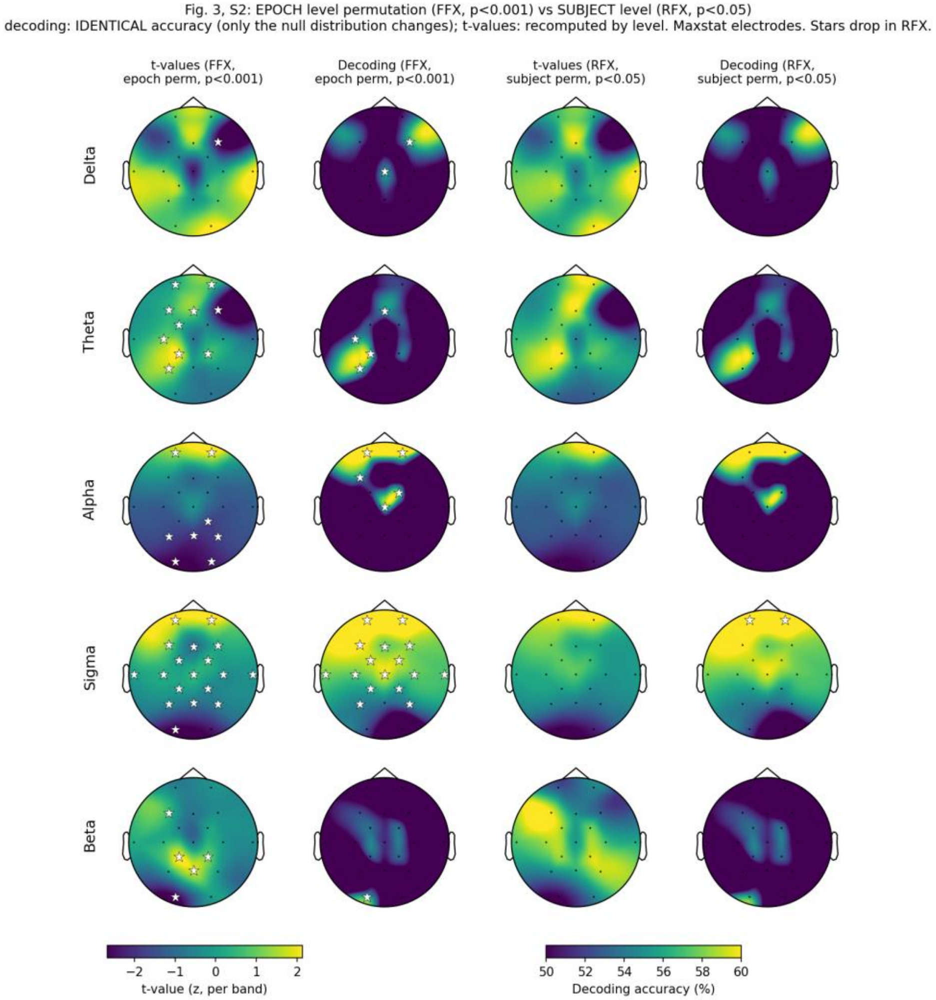


P-value survival curve by correction



T-test : Permutation Level: What Changes and What Doesn't





The inconsistency in Fig. 3 :
“t-values” panel: RFX, permutation at the subject level
“accuracies” panel: FFX, permutation at the epoch level
=> contrast that stems in part from the criterio

Approaches Explored and Discarded

Preprocessing:

- **ICA** (EOG rejection, threshold 0.6): accuracy drops to 44–54% compared to 54–69% in noica. ICA destroys discriminant signal. However, the threshold was set empirically using three converging methods.
- **ICLabel** (probability > 0.9), only 8% agreement with the EOG method; trained on wakefulness data and therefore unsuitable for sleep. Causes a drop in cosp_sigma/S2, our main result. Kept for comparison, not as part of the pipeline.
- **find_bads_muscle**, removed; no stable threshold range (CV 0.77–1.13 across the entire range); rejections confirmed as aberrant upon visual inspection.Z-score criterion for ICA, eliminated; 78.9% of subjects had zero rejections at z = 2.5; insufficient resolution with ~14 components per subject.
- **CAR** (Common Average Reference), removed; not recommended with fewer than ~64 electrodes; contaminates connectivity measurements and results in rank-deficient matrices.
- **AutoReject**,replaced by Potato.
- **And some others**

Approaches Explored and Discarded

Methods and Parameters :

- **FOOOF knee mode**, $\Delta R^2 = -0.0004$ across 139,000 spectra, 50× below the 0.02 threshold. Fixed value selected.
- **Linear subtraction for the oscillatory component**; values at ~1e-10, below the machine's epsilon, unusable. Ratio retained.
- **250 Hz**, under-resolution of the sigma band; abandoned in favor of 1,000 Hz.
- **EFS multi-feature**; extension beyond Arthur abandoned. Result: no super-combinations; redundant features; cosp_sigma dominates S2 and cosp_delta dominates SWS. Two issues specific to our version: the ~84% cross-state figure is an SPD aggregation artifact and not a reportable accuracy, and the p-value suffers from double-dipping (pool already preselected by maxstat, reselected at each permutation).
- **And some others**

TY everyone !!!

CocoLab